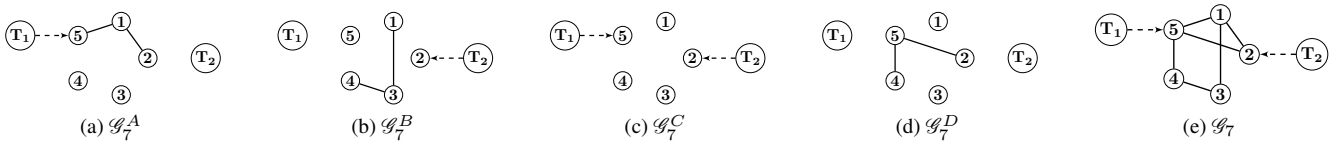
Supplementary

1 Effects of gain on the performance of proposed algorithm



(a) $\mathscr{G}_7^A$ (b) $\mathscr{G}_7^B$ (c) $\mathscr{G}_7^C$ (d) $\mathscr{G}_7^D$ (e) $\mathscr{G}_7$

Figure 1: Sensing network instances $\mathscr{G}_7^t \in \{\mathscr{G}_7^A, \mathscr{G}_7^B, \mathscr{G}_7^C, \mathscr{G}_7^D\}$ among 5 agents and 2 targets. Solid lines denote inter-agent subgraphs $\mathscr{G}_5^t$, while dotted lines indicate target-to-agent binary information flows, where $\mathscr{G}_7 \triangleq \bigcup_{k\in\{A,B,C,D\}} \mathscr{G}_7^k$.

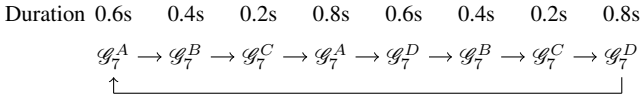
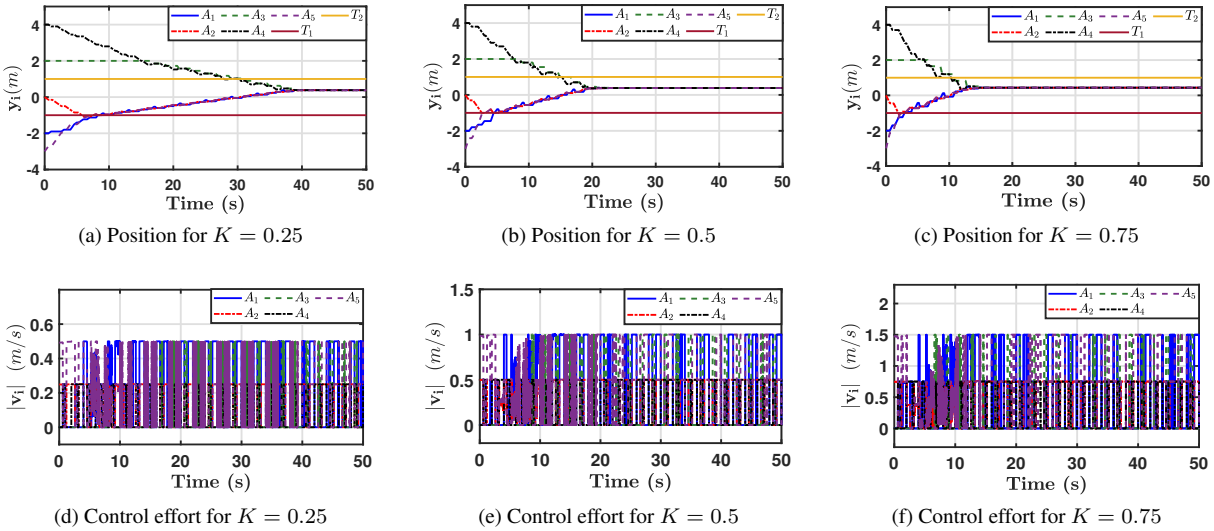


Figure 2: Network switching sequence showing minimum dwell time duration of 0.2 s and joint connectivity over $\kappa = 2$ s.

Agents & Targets	A_1	A_2	A_3	A_4	A_5	T_1	T_2
Position (m)	-2	0	2	4	-3	-1	1

Table 1: Initial states of agents and targets

The simulations of the proposed algorithm are carried out with the same initial states of the agents and targets as given in Table 1 (same as Table I in the main manuscript) for different values of control gain $K = \{0.25, 0.5, 0.75\}$ under the same inter-agent network shown in Fig. 1 (same as Fig. 1 in the main manuscript) with switching sequence given in Fig. 2 (same as Fig. 2 in the main manuscript). In Fig. 3 of this supplementary material, the trajectories of agents and targets are plotted for $K = \{0.25, 0.5, 0.75\}$ along with their respective control effort $|v_i|$. From these simulations, we observe that the convergence time decreases with increasing gain values. However, there is a trade-off for setting higher gain values, as it results in more control effort (refer to Fig. 3). The same observation is tabulated in Table 2 as well.



(a) Position for $K = 0.25$ (b) Position for $K = 0.5$ (c) Position for $K = 0.75$

(d) Control effort for $K = 0.25$ (e) Control effort for $K = 0.5$ (f) Control effort for $K = 0.75$

Figure 3: Performance of the proposed containment-consensus algorithm for $K = \{0.25, 0.5, 0.75\}$. Observation: Convergence time improves and control effort increases with increasing K values.

Control gain K	0.25	0.5	0.75
Convergence Time (s)	40	25	15
$\max_{\forall i}\{\sup_{\forall t\geq 0}\ \lvert v_i(t)\rvert\}$ **(m/s)**	0.5	1	1.5

Table 2: Effect of K on convergence time and control effort

2 Video description of simulator-in-the-loop (SITL) simulations

The SITL simulation is demonstrated in the **VIDEO** [1]. In this video, `crazyflie2` and `crazyflie5` correspond to targets T_1 and T_2, whereas `crazyflie1`, `crazyflie3`, `crazyflie4`, `crazyflie6`, and `crazyflie7` are assigned as agents A_1, A_2, A_3, A_4, and A_5, respectively.

From this video, the following are observed in the first 5 seconds ($0 \leq t \leq 5$ s):

- All nano-quadrotors take off to an altitude of $z^* = 1.5$ m and establish positions corresponding to the same initial y-coordinates as given in Table 1. The target nano-quadrotors `crazyflie2` and `crazyflie5` remain fixed at their initial coordinates to anchor the 1D convex hull along y-axis. The rest of the nano-quadrotors representing 5 agents are spawned and initialized across the workspace within this time frame, preparing to execute the proposed algorithm.

Execution and Convergence Analysis The progression observed in the SITL simulation highlights four key dynamic phases:

- **Initialization** ($5 \leq t < 25$ **s):** Follower nano-quadrotors (`crazyflie1, 3, 4, 6, 7`) begin from dispersed positions across the grid, evaluating binary sign measurements relative to active neighbors.
- **Transient Dynamic Compensation** ($25 \leq t < 70$ **s):** Under the switching sensing topology $\mathscr{G}_n^t$, agents publish velocity commands that reactively adjust trajectories without chattering, drawing closer to the target line segment.
- **Hull Containment** ($t \approx 72$ **s):** All five agents (A_1-A_5) contained within the target convex hull of `crazyflie2` (T_1) and `crazyflie5` (T_2).
- **Consensus Realization** ($t \approx 80$ **s):** The nano-quadrotors completes a straight collinear formation, converging precisely to the consensus position. Thus, the error dynamics $|y_i - \overline{y}|$ decay asymptotically to zero.

[1] `https://youtu.be/toub7Jv4VMU`